
V2AM – Vision to Actuation Mechanics

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Abstract

Existing Video Action Models rely on backbones with billions of parameters too slow for closed loop control, and their unconstrained action decoders emit jerky trajectories unsafe on real hardware. We present **V2AM**, a factorized framework that pairs a LoRA adapted Hummingbird image to video generator, a frozen VideoMAE encoder, and a compact flow matching action head implemented from scratch. We regularize the action head with a kinematic smoothness term $\mathcal{L}_{\text{smooth}}$ on the predicted velocity chunk, and at inference ensemble $K = 5$ noise seeds to deploy a mean trajectory and a per timestep epistemic uncertainty signal. On MimicGen, the baseline reaches $R^2 = 0.72$ and 95.9% gripper accuracy on coffee pod insertion; ensembling cuts validation MSE by 17.0% at no training cost; and $\mathcal{L}_{\text{smooth}}$ at $\lambda = 0.5$ cuts mean trajectory jerk by 38.5% with no accuracy loss, with $2.9\times$ larger reduction on contact rich cube stacking than on smooth coffee pod insertion.

Track: Methods. Physics informed regularization for generative robot policies.

1 Introduction

Robot manipulation requires anticipation: to grasp or insert, the policy must reason about where the scene will be in the next few hundred milliseconds, not only where it is now. Vision Language Action models such as π_0 (1) and OpenVLA (2) condition only on the current frame, infer physical causality from demonstrations alone, and still fail on novel object configurations. Video Action Models (VAMs) (3; 4; 5) close this gap by pairing a generative video backbone with an action decoder, but rely on 2B or larger parameter generators incompatible with closed loop control, and their unconstrained decoders routinely emit joint velocities that no actuator can track and discontinuous action chunks that would trigger hardware faults in practice.

We propose **V2AM**, a lightweight factorized framework that targets both failure modes at their source. We adapt the Hummingbird image to video model (7) with LoRA (13), reuse VideoMAE (8) pretrained on Something Something v2 as the frozen encoder, and implement a compact flow matching action head from scratch. Keeping each stage small lets every ablation fit on a single consumer GPU. We regularize the action head training loss with a kinematic smoothness term that penalizes jerk in the predicted velocity chunk, and at inference ensemble several noise seeds for both a mean trajectory and a per timestep epistemic uncertainty signal.

Contributions:

- A factorized video action pipeline (LoRA adapted Hummingbird, frozen VideoMAE, compact Conditional Flow Matching action head) implemented from scratch and benchmarked on three MimicGen tasks of varying contact richness.
- A kinematic smoothness loss $\mathcal{L}_{\text{smooth}}$ that cuts mean trajectory jerk by 38.5% at $\lambda = 0.5$ with no accuracy cost, with $2.9\times$ larger reduction on contact rich cube stacking than on smooth coffee pod insertion.

- A flow ensembling procedure that cuts validation MSE by 17.0% at zero training cost and surfaces a per timestep epistemic uncertainty signal whose range grows with task contact richness.

2 Related Work

VAMs and VLAs. CosmosPredict (3), Unified Video Action (4) and Mimic Video (5) jointly train a video generator and an action decoder inside a single 2B+ parameter model, achieving strong benchmarks but at latencies incompatible with real time control. π_0 (1) and OpenVLA (2) use pretrained vision and language backbones with generative action heads, including Conditional Flow Matching in π_0 , while Diffusion Policy (11) made the same case earlier using a denoising diffusion head. None of these models include explicit forward dynamics, so they must learn physical causality from demonstrations alone and require very large robot data corpora. V2AM factorizes the same two stages into independently trained components so each failure mode can be regularized in isolation.

Backbones and data. VideoMAE (8) shows that masked spatiotemporal autoencoding yields transferable, data efficient representations; pretraining on Something Something v2, a dataset of hand object interactions, makes it a strong frozen encoder for manipulation. MimicGen (9) scales human teleoperation to thousands of diverse demonstrations by spatially transforming object centric segments. We consume MimicGen directly and reuse VideoMAE without retraining the encoder.

Conditional Flow Matching and physics informed losses. CFM (10) regresses a velocity field along a straight line interpolation between noise and data, integrated at inference time as an ODE; most prior work uses fixed step forward Euler. We adopt the same objective and integration and concentrate our contribution on a training side smoothness regularizer and on noise side uncertainty quantification. Physics Informed Neural Networks (12) encode physical constraints as structured loss terms, and classical robot motion planning likewise penalizes acceleration and jerk inside trajectory optimization. Learned policies typically handle smoothness at deployment time via low pass filtering; V2AM instead bakes a kinematic jerk penalty directly into the flow matching training loss.

Video world models. A complementary line of work treats video diffusion models as general purpose simulators that policies can roll out against, such as Vid2World (6). These models are powerful at long horizon scene dynamics but typically require additional action conditioning and full closed loop rollouts to drive a controller. V2AM uses the video backbone differently: as a frozen, short horizon feature extractor over imagined futures, leaving the action head to learn the inverse dynamics directly from those features.

3 Methodology

V2AM cascades a generative video backbone with a flow matching action head, trained independently. Given a current observation o_t and language instruction ℓ , Hummingbird synthesizes a short window of future frames; a frozen VideoMAE encoder compresses them into spatiotemporal tokens h ; and the action head conditions on h and proprioception s_t to produce a 16 step action chunk through Conditional Flow Matching (Figure 1). Each block is a small specialized network trained in isolation, which lets us regularize the action head with a smoothness term and a noise ensemble without touching the video backbone and keeps every ablation trainable on a single consumer GPU. We detail each component next and tie everything back together with the dataset, hyperparameters, and end to end inference cycle in Sec. 3.3.

3.1 Conditional Flow Matching Action Head

The action head frames action chunk generation as a continuous time transport problem. Let $x_0 \in \mathbb{R}^{C \times D}$ be a clean action chunk ($C = 16$, $D = 6$ for end effector position and orientation deltas) and $x_1 \sim \mathcal{N}(0, I)$ Gaussian noise of the same shape. The linear interpolant $x_t = (1 - t)x_0 + tx_1$ for $t \in [0, 1]$ has constant target velocity field $u_t = x_1 - x_0$, regressed by the denoiser v_θ via the Conditional Flow Matching loss (10):

$$\mathcal{L}_{\text{flow}}(\theta) = \mathbb{E}_{t, x_0, x_1} \left[\|v_\theta(t, x_t, h, s_t) - (x_1 - x_0)\|^2 \right] \quad (1)$$

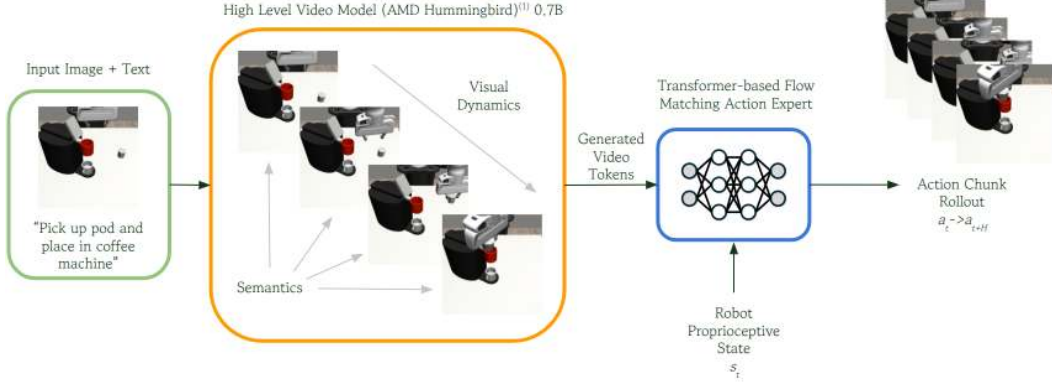


Figure 1: High level V2AM pipeline. The current camera frame o_t and language instruction ℓ condition the LoRA adapted Hummingbird video generator; a frozen VideoMAE encoder compresses the predicted future frames into feature tokens h , which the flow matching action head consumes together with proprioception s_t to emit a 16 step action chunk. A more detailed view is given in Fig. 4.

where $h \in \mathbb{R}^{1568 \times 768}$ are VideoMAE features pooled across two camera streams and $s_t \in \mathbb{R}^8$ is proprioception.

Architecture. The denoiser is a transformer with hidden dimension 384, four pre norm decoder blocks, and six attention heads (10.37M trainable parameters). Each block applies cross attention from noisy action tokens into the VideoMAE plus proprioception context, self attention among action tokens, and a feed forward MLP. A sinusoidal embedding of the integration time t is fused with the noisy action tokens via a small MLP before the first block. Two linear heads decode the final hidden states to a $C \times 6$ continuous velocity field and a $C \times 1$ binary gripper command trained with cross entropy $\mathcal{L}_{\text{grip}}$.

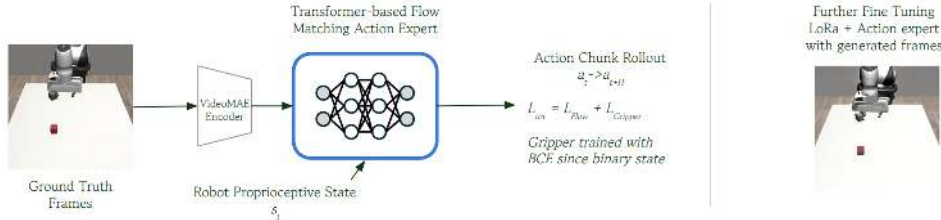


Figure 2: Action head training. Eight future frames at stride 2 from the agent view and wrist cameras pass through the frozen VideoMAE encoder to produce 1568 spatiotemporal tokens that condition the transformer flow matching denoiser together with proprioception s_t , predicting the velocity field for a 16 action chunk.

Training objective. The total loss combines $\mathcal{L}_{\text{flow}}$ (Eq. 1), the gripper cross entropy $\mathcal{L}_{\text{grip}}$, and a kinematic smoothness penalty on the predicted velocity chunk that encodes a bounded jerk constraint at the loss level:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{C-1} \sum_{k=1}^{C-1} \|v_{\theta}^{(k+1)} - v_{\theta}^{(k)}\|^2, \quad \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{grip}} + \lambda \mathcal{L}_{\text{smooth}} \quad (2)$$

where $v_{\theta}^{(k)}$ is the predicted velocity at chunk slot k and λ is swept over $\{0, 0.05, 0.1, 0.5\}$. This puts the smoothness prior on the trajectory generator itself rather than on a downstream low pass filter, in line with physics informed learning (12).

Inference. We draw $x_1 \sim \mathcal{N}(0, I)$ and integrate the field from $t = 1$ to $t = 0$ using $N = 50$ fixed step Euler updates,

$$x_{t+\Delta t} = x_t + \Delta t v_{\theta}(t, x_t, h, s_t), \quad \Delta t = -1/N. \quad (3)$$

We run this integration $K = 5$ times with independent noise seeds and deploy the empirical mean $\bar{x}_0 = \frac{1}{K} \sum_{i=1}^K x_0^{(i)}$ as the action chunk; the per timestep variance $\sigma_k^2 = \frac{1}{K} \sum_i \|x_0^{(i),k} - \bar{x}_0^k\|^2$ is logged as an epistemic uncertainty signal at zero additional training cost.

3.2 Generative Backbone Adaptation

Hummingbird (7) is a 0.7B parameter image to video diffusion model trained on internet video. We adapt it to the MimicGen domain with Low Rank Adaptation (13) (Figure 3) on the diffusion denoiser attention layers only; the variational autoencoder and text encoder stay frozen, leaving $\sim 2\text{M}$ parameters (0.1% of the model) trainable. Configuration and schedule are in Sec. 3.3.

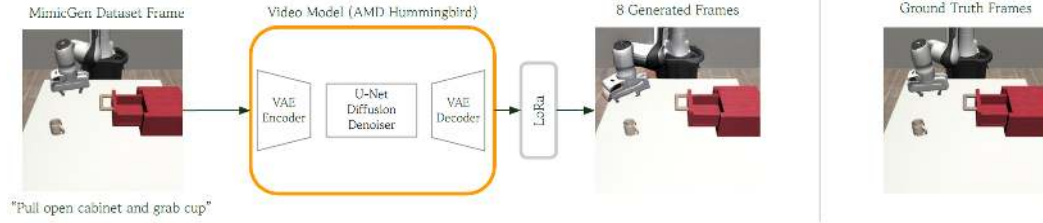


Figure 3: Hummingbird LoRA adaptation. Low rank updates are injected into the diffusion denoiser attention layers while the variational autoencoder and text encoder remain frozen; the model learns to generate eight future frames from the current observation and a language instruction.

3.3 Implementation Details

Dataset. We use MimicGen (9), which generates diverse robot demonstrations by spatially transforming human teleoperation traces. We focus on three tasks spanning a range of contact richness: coffee pod insertion (free motion), cube stacking (single grasp and place), and stack three cubes (multiple grasps). Episodes are LeRobot parquet files with two camera streams (agent view and wrist), proprioception, and seven dimensional actions. Windows are sampled at action stride one and frame stride two, producing 8 frames per clip and 16 timesteps per action chunk; we use ~ 40 training and ~ 10 validation episodes per task. Continuous actions are z score normalized per dataset; the gripper command is mapped to $\{0, 1\}$.

Training pipeline. Figure 4 shows the data flow at deployment; training proceeds in two independent stages.

Stage 1: Hummingbird LoRA adaptation. Low Rank Adaptation modules are injected into the diffusion denoiser attention layers while the VAE and text encoder stay frozen. We train on eight frame clips at stride two, resolution 256×256 , with a four frame sliding window for temporal consistency, on a single NVIDIA RTX 5060 Ti with gradient accumulation (effective batch size eight). Table 1 lists the configuration.

Table 1: Hummingbird LoRA training configuration.

Hyperparameter	Value	Hyperparameter	Value
LoRA rank (r)	8	Resolution	256×256
LoRA α	32	Video length	8 frames
LoRA dropout	0.05	Frame stride	2
Learning rate	1×10^{-4}	Sliding window	4 frames
Batch size	1 (effective 8)	Precision	mixed FP16
Max epochs	50	Hardware	RTX 5060 Ti

Stage 2: Action head training. The frozen VideoMAE encoder produces $h \in \mathbb{R}^{1568 \times 768}$ from ground truth future frames, concatenated with proprioception and fed to the transformer denoiser. The action head trains from scratch with $\mathcal{L}_{\text{total}}$ (Eq. 2); the encoder and LoRA adapted generator stay strictly frozen, so only the 10.37M action head parameters update. Table 2 lists the configuration.

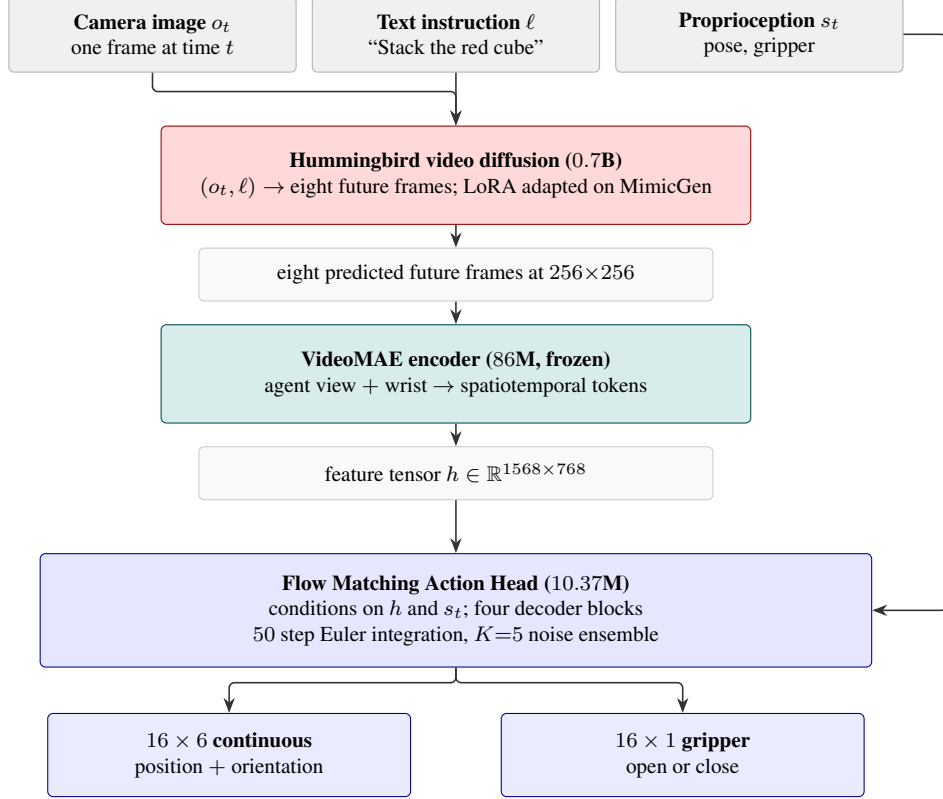


Figure 4: V2AM end to end architecture. Camera frame o_t and instruction ℓ enter the LoRA adapted Hummingbird video diffusion model, which synthesizes eight future frames; together with the wrist view, they are encoded by frozen VideoMAE into $h \in \mathbb{R}^{1568 \times 768}$. Proprioception s_t bypasses the visual stack and joins h inside the flow matching action head, which integrates from Gaussian noise to clean actions in 50 Euler steps with a $K = 5$ noise ensemble and emits the 16 step chunk plus binary gripper commands.

Inference pipeline. Given o_t , ℓ , and s_t , one cycle proceeds as follows: (i) Hummingbird generates eight predicted future agent view frames (wrist view taken from o_t); (ii) the two camera eight frame clip is encoded by VideoMAE to produce h ; (iii) we draw $K = 5$ noise samples $x_1^{(i)} \sim \mathcal{N}(0, I)$ and run 50 step forward Euler integration of the velocity field conditioned on h and s_t for each; (iv) the empirical mean is deployed as the 16 step action chunk and the per timestep variance is logged. The robot executes the chunk open loop and the cycle repeats.

4 Results & Analysis

4.1 Experimental Setup

We evaluate on three MimicGen tasks spanning contact richness: coffee pod insertion (free motion), cube stacking (single grasp), and stack three cubes (multiple grasps). Validation uses the held out parquet split (186 windows of 16 timesteps on coffee pod, comparable counts on the others). We track validation MSE, R^2 , gripper accuracy, mean trajectory jerk (average squared difference between consecutive predicted continuous actions), per timestep variance across the $K = 5$ noise ensemble, and end to end task success in closed loop robosuite/MuJoCo rollouts. Training and inference settings follow Sec. 3.3; checkpoint selection uses 50 step Euler validation MSE unless noted.

Table 2: Action head training and inference configuration.

Component	Configuration
Visual encoder	Frozen VideoMAE (ViT base, 6th layer)
Input sequence	1568 spatiotemporal tokens (dual camera)
Denoiser architecture	4 transformer blocks, 6 heads, hidden dim 384
Trainable parameters	10.37M (action head only)
Action chunk	16 timesteps, 6 continuous + 1 gripper
Optimizer	AdamW
Learning rate	1×10^{-4} , 5 epoch warmup, cosine schedule
Weight decay	1×10^{-4}
Batch size	32
Total epochs	100
Gradient clip	1.0
Smoothness weight λ	swept over $\{0, 0.05, 0.1, 0.5\}$
Inference solver	50 step forward Euler
Noise ensemble	$K = 5$ seeds

4.2 Generative Backbone Fidelity

We verify how faithfully the LoRA adapted Hummingbird model reproduces ground truth video on held out clips: five episodes per task, ten conditioning frames per episode, eight frame rollouts at stride two, compared per frame against ground truth (Table 3). Mean PSNR across the six tasks is 19.33 ± 1.07 , SSIM 0.8561 ± 0.0246 , LPIPS 0.1090 ± 0.0258 . The model is most faithful on continuous motion (coffee pod: PSNR 20.31, LPIPS 0.077) and least faithful on fine spatial geometry (square nut on peg: PSNR 17.28, LPIPS 0.157), matching the expected failure mode of a small video diffusion model.

Table 3: Hummingbird LoRA generation fidelity across six MimicGen tasks (5 episodes per task, 10 conditioning frames per episode, 8 predicted frames per clip at stride 2). PSNR and SSIM are higher is better; LPIPS is lower is better.

Task	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Coffee pod insertion	20.31 ± 0.46	0.8788 ± 0.0047	0.0765 ± 0.0038
Mug cleanup	19.82 ± 0.31	0.8552 ± 0.0060	0.1040 ± 0.0071
Square nut on peg	17.28 ± 0.22	0.8073 ± 0.0153	0.1567 ± 0.0101
Cube stacking	19.39 ± 0.49	0.8677 ± 0.0096	0.1121 ± 0.0104
Needle threading	19.09 ± 0.32	0.8629 ± 0.0068	0.1119 ± 0.0044
Three piece assembly	20.10 ± 0.38	0.8647 ± 0.0075	0.0925 ± 0.0083
All tasks (mean)	19.33 ± 1.07	0.8561 ± 0.0246	0.1090 ± 0.0258

4.3 Value of Future Frames

We test whether imagined future frames help the action decoder on coffee pod insertion: same architecture trained with eight ground truth future frames (V2AM setting) versus the current frame replicated eight times (Table 4). Future frames raise R^2 from 0.615 to 0.723 (+0.108), lower MSE from 0.061 to 0.052 (−15%), and improve gripper accuracy from 94.8% to 95.9%. The gap justifies the additional Hummingbird inference cost: a policy that cannot see ahead cannot anticipate contact moments.

4.4 Flow Ensembling at Inference

Sampling $K = 5$ noise seeds and deploying the empirical mean cuts validation MSE from 0.05396 to 0.04481 on the $\lambda = 0$ baseline (−17.0%), lifts R^2 from 0.7173 to 0.7652, and drops mean trajectory jerk from 0.07780 to 0.01710 (−78%), all at zero training cost (Table 5). Jerk benefits disproportionately because it is a second order finite difference, which amplifies the high frequency

Table 4: Action head accuracy with and without future frame conditioning on coffee pod insertion. Future frames raise R^2 by +0.108 and cut MSE by 15%.

Input setup	Loss	Val MSE ↓	R^2 ↑	Gripper acc ↑
With future frames	1.06	0.052	0.723	95.9%
Current frame only	1.11	0.061	0.615	94.8%

noise that ensembling cancels. Gripper accuracy is essentially unchanged, indicating the discrete gripper logits already saturate. The $K = 5$ multiplier on solver evaluations ($10 \rightarrow 50$ network calls per chunk) remains compatible with 20 Hz closed loop control.

Table 5: Inference time effect of flow ensembling on the $\lambda = 0$ baseline (coffee pod). Ensembling alone cuts MSE by 17.0% and jerk by roughly an order of magnitude at zero training cost.

Inference	Val MSE ↓	R^2 ↑	Gripper acc ↑	Mean jerk ↓
Euler ($N = 10$)	0.05396	0.7173	95.87%	0.07780
Euler + ensemble ($K = 5$)	0.04481	0.7652	95.83%	0.01710
Relative change	−17.0%	+0.048	−0.04pp	−78.0%

4.5 Smoothness Regularization

Control: the gain is from $\mathcal{L}_{\text{smooth}}$, not from checkpoint selection. The smoothness experiments use 50 step Euler validation MSE for checkpoint selection while the original baseline used 10 step. To isolate the regularizer from the criterion, we retrained at $\lambda = 0$ with 50 step Euler checkpoint selection (Table 6): MSE shifts by only +0.4%, at the noise floor. Turning $\mathcal{L}_{\text{smooth}}$ on at $\lambda = 0.1$ on top then brings MSE down to 0.05284, −2.4% vs control and −2.1% vs the original baseline. The gain is therefore the smoothness loss itself.

Table 6: Control isolating the smoothness effect from checkpoint selection on coffee pod insertion. Switching the validation criterion from 10 step to 50 step Euler shifts MSE by only +0.4%; $\mathcal{L}_{\text{smooth}}$ at $\lambda = 0.1$ on top of that adds a further −2.4%.

Configuration	Val MSE	Mean jerk	vs reference
$\lambda = 0$, checkpoint by 10 step Euler	0.05396	0.07780	reference
$\lambda = 0$, checkpoint by 50 step Euler	0.05415	0.06901	MSE +0.4%
$\lambda = 0.1$, checkpoint by 50 step Euler	0.05284	0.06586	MSE −2.4%

Lambda sweep. We sweep $\lambda \in \{0, 0.05, 0.1, 0.5\}$ on coffee pod with all other settings fixed (Table 7). Performance improves monotonically with no accuracy regression: MSE drops by −7.3%, R^2 rises from 0.7163 to 0.7371, gripper stays within 0.1pp, and mean Euler jerk falls −38.5% at $\lambda = 0.5$. Jerk reduction is non linear (−3.6%, −4.6%, −38.5% at $\lambda = 0.05, 0.1, 0.5$), with a sharp transition between $\lambda = 0.1$ and 0.5 consistent with $\mathcal{L}_{\text{smooth}}$ and $\mathcal{L}_{\text{flow}}$ being structurally non competing (one acts on the clean action chunk, the other on the velocity field). The best configuration is $\lambda = 0.5$ with $K = 5$ ensembling: MSE 0.04438, $R^2 = 0.7675$, jerk 0.00885.

Task amplification: contact richness matters. Cube stacking shows larger jerk reductions than coffee pod at every λ . At $\lambda = 0.1$: −13.3% on cube stacking versus −4.6% on coffee pod, a $2.9\times$ amplification; MSE scales similarly (−3.4% vs −2.4%). $\mathcal{L}_{\text{smooth}}$ targets kinematic jerk, so tasks with more naturally occurring jerk (discrete contact events) give it more raw signal to remove.

4.6 Uncertainty Quantification via Ensemble Variance

The per timestep variance across $K = 5$ noise samples is a free epistemic uncertainty signal. Two effects are visible. First, mean variance on coffee pod collapses with λ from 0.376 to 0.181 (−52%) between $\lambda = 0$ and $\lambda = 0.5$: the model becomes more deterministic across noise seeds as smoothness

Table 7: Lambda sweep on coffee pod with 50 step Euler checkpoint selection. Every metric improves monotonically with λ from 0 to 0.5 with no accuracy regression. Best configuration in bold.

λ	Euler MSE	R^2	Gripper	Euler jerk	Ensemble jerk	Jerk Δ vs $\lambda=0$
0.0	0.05415	0.7163	95.70%	0.06901	0.01519	reference
0.05	0.05348	0.7198	95.70%	0.06656	0.01283	−3.6%
0.1	0.05284	0.7232	95.77%	0.06586	0.01265	−4.6%
0.5	0.05018	0.7371	95.70%	0.04245	0.00885	−38.5%

takes hold. Second, contact rich tasks expose a larger *range* of variance across the 16 chunk timesteps; at $\lambda = 0.1$ cube stacking has range 0.069, more than twice the coffee pod range of 0.034. The variance range can serve as an implicit contact event indicator at deployment, at zero training cost.

4.7 End to End Task Success in Simulation

We close the loop with rollouts in *robosuite*/MuJoCo (Table 8). In the Oracle setting (action head fed ground truth future frames), the policy succeeds on 65.0% of coffee pod and 82.0% of cube stacking rollouts. With the full V2AM pipeline feeding Hummingbird generated frames, cube stacking success drops to $\sim 60\%$; the remaining gap is the simulation to generated frame domain shift, mitigated but not eliminated by fine tuning the action head on cached Hummingbird rollouts. End to end success confirms what the per step metrics suggest, with the largest headroom on contact rich tasks under the full generative pipeline.

Table 8: End to end task success rate in closed loop *robosuite*/MuJoCo rollouts. The Oracle setting feeds ground truth future frames; the Full V2AM setting feeds Hummingbird generated frames into the same action head.

Setting	Coffee pod	Cube stack
Oracle (ground truth future frames)	65.0%	82.0%
Full V2AM (Hummingbird generated frames)	n/a	$\sim 60\%$

5 Conclusion

V2AM factorizes a robot manipulation policy into a LoRA adapted Hummingbird video generator, a frozen VideoMAE encoder, and a compact flow matching action head, and adds two physics informed components: a kinematic smoothness loss $\mathcal{L}_{\text{smooth}}$ on the predicted velocity chunk and a noise seed ensemble at inference. On MimicGen, ensembling alone cuts validation MSE by 17.0% at zero training cost, and $\mathcal{L}_{\text{smooth}}$ at $\lambda = 0.5$ reduces mean trajectory jerk by 38.5% with no accuracy loss, with $2.9\times$ larger reduction on contact rich cube stacking than on smooth coffee pod insertion. Future work includes pushing the smoothness sweep beyond $\lambda = 0.5$, real robot deployment, and integrating the per timestep variance signal into downstream replanning.

6 Code & GitHub

The full code and evaluation scripts are available at the following [link](#).

7 Team Member Contribution

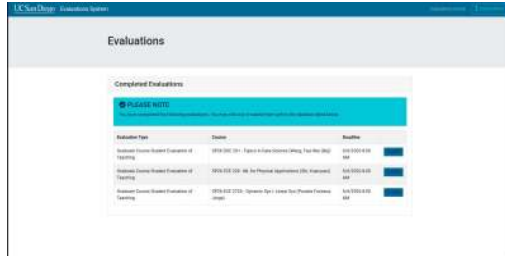
Shivam Yadav handled literature review, writing, and presentation. Siddharth Rama Sushil integrated the physics informed components into the architecture and ran all ablation studies, inferences, and results. Shahid Mohammad Mulla and Sri Mihir Ungarala built the action head architecture from scratch. Overall, all members contributed equally to the project.

References

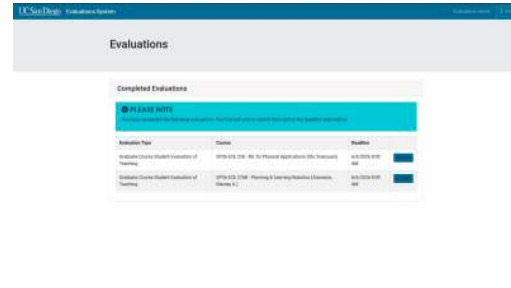
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Bonus: Course Evaluation Submission

The course evaluation has been completed by every member of the team. Screenshots of each submission are provided below.



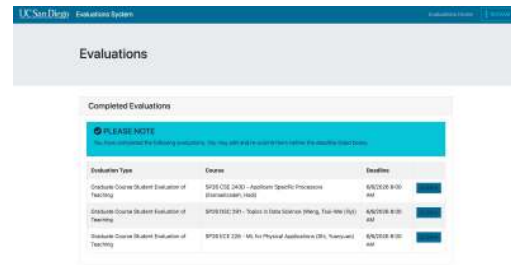
(a) Siddharth Rama Sushil.



(b) Sri Mihir Ungarala.



(c) Shahid Mohammad Mulla.



(d) Shivam Yadav.

Figure 5: Course evaluation submissions from all team members.